

INFECTION CONTROL

NCLEX STUDY GUIDE

Asepsis • Isolation • PPE • Communicable Diseases

1. Chain of Infection and Asepsis

Link	Meaning / nursing control
Agent	Bacterium, virus, fungus or other pathogen.
Reservoir	Place where pathogen survives/grows.
Portal of exit	Cough, sneeze, saliva, blood, drainage or excreta.
Transmission	Contact, droplet, airborne or contaminated vehicle; nursing precautions interrupt this link.
Portal of entry	Eyes, nose, mouth, broken skin or invasive device.
Susceptible host	Person at risk because of age, illness, immunity or treatment.

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Medical asepsis	Surgical asepsis
Clean technique: hand hygiene, clean gloves/PPE; hands rinsed downward. Soap/water when visibly dirty and always for C. difficile.	Sterile technique: all field items sterile; outer 1 inch contaminated; wet field contaminated; keep field above waist/in view; open far flap first; never turn back.

2. Standard Precautions, PPE and Sharps

- Use for every patient and all blood/body fluids except sweat, nonintact skin and mucosa. Perform hand hygiene before and after contact.
- Don PPE: gown, mask/respirator, goggles/face shield, gloves. Remove the most contaminated first; perform hand hygiene.
- N95 respirator for airborne rooms; surgical mask for droplet exposure and for isolated patients during transport.
- Never recap a needle unless unavoidable; then use one-hand scoop. Dispose in puncture-resistant container before three-quarters full. Clean spills with approved disinfectant/bleach dilution per policy.

C. DIFFICILE Use contact precautions and wash hands with soap and water; alcohol sanitizer does not reliably remove spores.

3. Transmission-Based Precautions

Contact	Droplet	Airborne
Gown + gloves, dedicated equipment, door may remain open. Examples: diarrheal illness/C. diff, RSV, MRSA/VRSA, impetigo, conjunctivitis, scabies, lice and localized zoster.	Surgical mask near patient; private/cohort room, door open; patient wears mask for transport. Examples: influenza, pertussis, diphtheria, rubella, mumps, bacterial meningitis and pneumococcal disease.	Negative-pressure room, door closed, N95 and 6-12 air exchanges/hr. Examples: TB, measles, varicella/smallpox.

MULTIPLE PRECAUTIONS Disseminated herpes zoster: airborne + contact + standard. Follow current facility/CDC guidance for emerging pathogens such as MERS, SARS and viral hemorrhagic fever.

4. Vector-Borne and Fungal Infections



Lyme disease: erythema migrans or bull's-eye lesion

- Lyme disease: deer-tick *Borrelia* infection. Remove tick with tweezers without squeezing; save/seal if required. Early flu-like illness and bull's-eye rash; later Bell palsy/cardiac findings and chronic arthritis/fatigue. Treat promptly with doxycycline/other antibiotic as ordered.
- Rocky Mountain spotted fever: tick exposure with fever/headache/myalgia and rash on wrists/ankles, palms/soles; use protective clothing, repellent and yard/tick control.



Tinea corporis (ringworm): fungal annular lesion

- Tinea corporis/pedis is fungal, not bacterial. Keep skin/feet dry and use antifungal such as clotrimazole, miconazole or ketoconazole; monitor liver toxicity with systemic therapy.
- Infectious mononucleosis: Epstein-Barr virus, fatigue, sore throat, lymphadenopathy and splenic enlargement. Standard precautions; avoid sharing utensils/kissing and contact sports. Left-shoulder pain may indicate splenic rupture.

5. Contact Diseases



Impetigo: characteristic honey-colored crust around the mouth

Disease	Key care
Impetigo	Contact until 24-48 hr after effective antibiotics; gentle cleansing/warm compress, topical antibiotic, separate linens. Watch edema or new murmur for post-strep complications.
Pediculosis	Permethrin and nit comb; boil/discard combs, hot-wash items, seal nonwashables about 2 weeks and vacuum. Nits alone do not indicate active transmission.
Scabies	Nocturnal pruritus and burrows in finger webs/wrists/axilla/groin; permethrin overnight, hot wash/dry, seal items and repeat treatment. Lindane contraindicated in young children/pregnancy.

6. Droplet Diseases

Disease	Hallmark / priority
Rubella	Pink rash face-to-body and posterior/occipital nodes; contagious 7 days before/after rash. Avoid pregnant contacts. Postpartum MMR when nonimmune; avoid pregnancy afterward per current guidance.
Pertussis	Catarrhal stage then paroxysmal whoop/vomiting. Droplet until effective antibiotic period; humidified oxygen/suction and reduce irritants.
Diphtheria	Gray pseudomembrane, foul discharge and “bull neck”; airway equipment, antitoxin and antibiotics.
Scarlet fever	Strep: strawberry tongue, sandpaper rash and Pastia lines. Watch glomerulonephritis or rheumatic fever.
Mumps	Parotid pain/swelling, worse with sour foods; orchitis/infertility and meningitis risk. Scrotal support and compresses.
Parvovirus B19	Slapped-cheek rash; no longer contagious once rash appears. Avoid pregnant contacts.

7. Airborne Diseases

Disease	Key findings / care
Measles (rubeola)	3 Cs: cough, coryza, conjunctivitis; Koplik spots and descending rash. Airborne; cool mist, dim light for photophobia and no aspirin.
Varicella	Lesions in multiple stages simultaneously; airborne + contact until all crusted. Calamine/oatmeal, antihistamine, mittens/trim nails and no aspirin.
Herpes zoster	Dermatomal burning/allodynia; localized contact, disseminated airborne + contact. Moist dressings, early antiviral and neuropathic pain drug; urgent eye evaluation for facial/ocular disease.
Tuberculosis	Airborne and N95 for all entering. Continue isolation until criteria met per public-health/facility policy.

8. Emerging and High-Consequence Infections

- MERS/SARS: severe coronavirus respiratory disease with fever, cough, dyspnea and respiratory failure; use current enhanced precautions.
- Ebola/viral hemorrhagic fever: fever, petechiae and hemorrhage; strict PPE, contact/droplet and added airborne protection for aerosol procedures per protocol.

9. Biological and Chemical Agents

Agent	Hallmark / response
Anthrax	Not usually person-to-person. Inhaled: severe respiratory illness/hemoptysis; cutaneous: painless black eschar; GI: bleeding. Decontaminate and treat with ciprofloxacin/doxycycline.
Botulism	Home-canned food/honey in infant; descending paralysis, ptosis, dysphagia and respiratory failure. Airway and antitoxin.
Plague	Bubonic nodes, septicemia or pneumonic cough; pneumonic form requires droplet. Streptomycin/gentamicin.
Smallpox	High fever and dense facial rash; airborne/contact and public-health emergency.
Tularemia	Rabbit/tick exposure, fever, weakness and pneumonia; aminoglycoside/doxycycline.
Mustard gas	Pain-free blisters, eye injury and respiratory burns; decontaminate/supportive care.
Sarin	Cholinergic SLUDGE findings, seizure and paralysis; atropine + pralidoxime.

10. Immunization Review

1	2	4	6	12 15 18	19-23 2 4-6 11-12 16
Birth		M	onths		Years
Hep B	Hep B			Hep B	
#1	#2			#3	
RV	RV	RV*			
#1	#2	#3			
DTaP	DTaP	DTaP	DTaP		DTaP
#1	#2	#3	#4		#5

Vaccination schedule - birth through early infancy

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Hib	Hib	Hib	Hib	
#1	#2	#3	#4	
PCV	PCV	PCV	PCV	
#1	#2	#3	#4	
IPV	IPV		IPV	IPV
#1	#2		#3	#4
			Hep A	
			#1&3	
			MMR	MMR

Vaccination schedule - later infancy and childhood

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			#1	#2
			VZV	VZV
			#1	#2
				Tdap
				#1
				HPV**
				#1,2
				MCV MCV
				#1 #2

Vaccination schedule - childhood and adolescent vaccines

Vaccination Abbreviations

Abbreviation	Meaning
Hep B	Hepatitis B vaccine
RV	Rotavirus vaccine: 2 doses when Rotarix is used; 3 doses when RotaTeq is used
DTaP	Diphtheria and tetanus toxoids with acellular pertussis vaccine
Tdap	Tetanus toxoid, reduced diphtheria toxoid and acellular pertussis vaccine
Hib	Haemophilus influenzae type b conjugate vaccine
PCV	Pneumococcal conjugate vaccine
MenB	Serogroup B meningococcal vaccine
IPV	Inactivated poliovirus vaccine
MMR	Measles, mumps and rubella vaccine
VZV	Varicella-zoster virus vaccine
Hep A	Hepatitis A vaccine
MCV	Meningococcal conjugate vaccine
HPV	Human papillomavirus vaccine; 3-dose series when initiated at age 15 or older

- Verify current CDC schedule, contraindications, pregnancy status and immune suppression before live vaccines.
- Vaccines for hepatitis A and B prevent those infections; hepatitis C/D/E do not have routine U.S. vaccines (HBV vaccination prevents hepatitis D coinfection).
- Document vaccine, lot, site, route, date and education; observe and manage allergic reactions.

FINAL NCLEX FOCUS Select isolation by route, protect the airway first, perform correct hand hygiene/PPE, prevent sharps injury and report neurologic, respiratory or hemorrhagic deterioration immediately.